Please check the examination details below before entering your candidate information

Candidate surname	Other names

Centre Number | Candidate Number

Pearson Edexcel Level 3 GCE

Friday 16 May 2025

Afternoon

Paper reference **8FM0/23**

Further Mathematics

Advanced Subsidiary
Further Mathematics options
23: Further Statistics 1
(Part of options B, E, F and G)

You must have:
Mathematical Formulae and Statistical Tables (Green), calculator

Total Marks

Candidates may use any calculator allowed by Pearson regulations. Calculators must not have the facility for symbolic algebra manipulation, differentiation and integration, or have retrievable mathematical formulae stored in them.

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- Inexact answers should be given to three significant figures unless otherwise stated.

Information

- A booklet 'Mathematical Formulae and Statistical Tables' is provided.
- The total mark for this part of the examination is 40. There are 4 questions.
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Advice

- Read each question carefully before you start to answer it.
- Try to answer every question.
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Turn over

Pearson

1. A researcher is investigating the relationship between a person's age and their preferred method of shopping.

A random sample of 300 people is taken and the results are summarised in the table below.

		Preferred method of shopping		
		Online	**In-store**	**Total**
Age	**18 – 30**	27	23	50
	31 – 50	31	59	90
	51 – 64	17	43	60
	65 and older	30	70	100
	Total	105	195	300

(a) Write down suitable hypotheses for a test to determine whether or not there is evidence of an association between age and preferred method of shopping.

(1)

The researcher assumes that the null hypothesis is true and uses the data in the table to find expected values.

(b) (i) Identify the cell which would have the smallest expected value.

(ii) Calculate the expected value for this cell.

(2)

The value of $\sum \frac{(O-E)^2}{E}$ for the other 7 cells is 5.060

(c) Test, at the 5% level of significance, whether or not there is evidence of an association between age and preferred method of shopping.
You should state the test statistic, degrees of freedom, critical value and conclusion clearly.

(5)

(d) Explain whether or not your conclusion to part (c) would be the same if the test was carried out at a 1% level of significance.

(2)

Question 1 continued

Question 1 continued

Question 1 continued

(Total for Question 1 is 10 marks)

2. The discrete random variable X represents the score when a spinner is spun.
The probability distribution of X is given by

x	2	5	9
$P(X = x)$	0.6	0.3	0.1

(a) Find Var(X)
Show your working clearly.
(4)

A game is played by spinning the spinner twice.

If the two scores are the same, the number of **points** earned is 0

If the two scores are different, the number of **points** earned is the sum of the two scores.

(b) Show that the probability of earning 14 **points** in one game is 0.06
(1)

(c) Find the expected number of **points** earned when the game is played once.
(4)

Mehmet plays the game 150 times.

(d) Using a Poisson approximation, find the probability that Mehmet earns 14 **points** in exactly 4 of the games.
(3)

Question 2 continued

Question 2 continued

Question 2 continued

(Total for Question 2 is 12 marks)

3. Raoul, Steffi and Taro are catching butterflies for research.

The number of butterflies caught by Raoul per hour may be assumed to follow a Poisson distribution with mean 4.2

Find the probability that Raoul catches

(a) (i) exactly 6 butterflies in a randomly selected one-hour period,

(1)

(ii) exactly 1 butterfly in a randomly selected 10-minute period.

(2)

Following a long period without rain, Raoul believes there will now be a change to the rate at which he catches butterflies.
To test his belief, he uses the random variable R to represent the number of butterflies he catches in a 4-hour period.

A hypothesis test is to be carried out to determine whether or not there is support for Raoul's belief. The null hypothesis of the test will be rejected if $R \leqslant 9$ or $R > 25$

(b) Stating the hypotheses clearly, find the actual level of significance of the test.

(4)

The number of butterflies caught by Steffi per hour may be assumed to follow a Poisson distribution with mean 3.2

(c) Find the probability that in exactly 2 of the next 4 hours, Steffi catches less than or equal to 3 butterflies each hour.

(3)

The number of butterflies caught by Taro per hour may be assumed to follow a Poisson distribution with mean 2.7

Steffi and Taro both go to catch butterflies in a field one day.

Taro models the **total** number of butterflies caught per hour with a Poisson distribution with mean 3.2 + 2.7 = 5.9

(d) State a condition that would be needed for Taro's model to be valid.

(1)

Question 3 continued

Question 3 continued

Question 3 continued

(Total for Question 3 is 11 marks)

4. A 5-question test is taken by 80 students.

Each question was answered either correctly or incorrectly.

Number of questions answered correctly	0	1	2	3	4	5
Observed frequency	15	23	31	9	2	0

(a) Show that the proportion of questions answered correctly by these 80 students is 0.3 **(2)**

Anisa believes that the number of questions answered correctly can be modelled using a binomial distribution.
In order to test her belief, Anisa first calculates some of the expected frequencies.

Number of questions answered correctly	0	1	2	3	4	5
Expected frequency	13.45	r	s	10.58	2.27	0.19

(b) Find the value of r and the value of s **(3)**

(c) Explain fully why there are 2 degrees of freedom for the test of Anisa's belief. **(2)**

Question 4 continued

Question 4 continued

(Total for Question 4 is 7 marks)

TOTAL FOR FURTHER STATISTICS 1 IS 40 MARKS

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Friday 17 May 2024

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Paper reference **8FM0/23**

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Advanced Subsidiary
Further Mathematics options
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Total Marks

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Turn over

1. Sharma believes that each computer game he sells appeals equally to all age ranges.
To investigate this, he takes a random sample of 100 people who play these games and asks them which of the games A, B or C they prefer.
The results are summarised in the table below.

Computer game		A	B	C
Age range	< 20	8	15	6
	$20 – 30$	21	12	9
	> 30	6	10	13

(a) Write down hypotheses for a suitable test to assess Sharma's belief.
(1)

(b) For the test, calculate the expected frequency for

(i) those players aged under 20 who prefer game C

(ii) those players aged between 20 and 30 who prefer game A
(2)

(c) State the degrees of freedom of the test statistic for this test.
(1)

Sharma correctly calculates the test statistic for this test to be 11.542 (to 3 decimal places).

(d) Using a 5% significance level, and stating your critical value, comment on Sharma's belief.
(2)

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Question 1 continued

Question 1 continued

Question 1 continued

(Total for Question 1 is 6 marks)

2. A manager keeps a record of accidents in a canteen.

Accidents occur randomly with an average of 2.7 per month. The manager decides to model the number of accidents with a Poisson distribution.

(a) Give a reason why a Poisson distribution could be a suitable model in this situation. **(1)**

(b) Assuming that a Poisson model is suitable, find the probability of

(i) at least 3 accidents in the next month, **(1)**

(ii) no more than 10 accidents in a 3-month period, **(2)**

(iii) at least 2 months with no accidents in an 8-month period. **(4)**

One day, two members of staff bump into each other in the canteen and each report the accident to the manager. The canteen manager is unsure whether to record this as one or two accidents.

Given that the manager still wants to model the number of accidents per month with a Poisson distribution,

(c) state

- a property of the Poisson distribution that the manager should consider when deciding how to record this situation
- whether the manager should record this as one or two accidents

(1)

The manager introduces some new procedures to try and reduce the average number of accidents per month.

During the following 12 months the total number of accidents is 22
The manager claims that the accident rate has been reduced.

(d) Use a 5% level of significance to carry out a suitable test to assess the manager's claim.
You should state your hypotheses clearly and the *p*-value used in your test. **(4)**

Question 2 continued

Question 2 continued

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Question 2 continued

(Total for Question 2 is 13 marks)

3. The discrete random variable X has probability distribution,

x	-1	0	1	3	7
$\mathrm{P}(X=x)$	p	r	p	0.3	r

where p and r are probabilities.

Given that $\mathrm{E}(X) = 1.95$

find the exact value of $\mathrm{E}\left(\sqrt{X+1}\right)$ giving your answer in the form $a+b\sqrt{2}$ where a and b are rational.

(6)

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Question 3 continued

(Total for Question 3 is 6 marks)

P 7 5 6 7 3 A 0 1 1 1 6

Turn over

4. Robin shoots 8 arrows at a target each day for 100 days.

The number of times he hits the target each day is summarised in the table below.

Number of hits	0	1	2	3	4	5	6	7	8
Frequency	1	10	30	34	17	4	2	0	2

Misha believes that these data can be modelled by a binomial distribution.

(a) State, in context, two assumptions that are implied by the use of this model.

(2)

(b) Find an estimate for the proportion of arrows Robin shoots that hit the target.

(2)

Misha calculates expected frequencies, to 2 decimal places, as follows.

Number of hits	0	1	2	3	4	5	6	7	8
Expected frequency	2.81	12.67	r	28.05	19.73	s	2.50	0.40	0.03

(c) Find the value of r and the value of s

(3)

Misha correctly used a suitable test to assess her belief.

(d) (i) Explain why she used a test with 3 degrees of freedom.

(2)

(ii) Complete the test using a 5% level of significance.
You should clearly state your hypotheses, test statistic, critical value and conclusion.

(6)

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Question 4 continued

Question 4 continued

Question 4 continued

Question 4 continued

(Total for Question 4 is 15 marks)

TOTAL FOR FURTHER STATISTICS 1 IS 40 MARKS

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Friday 19 May 2023

Afternoon

Paper reference **8FM0/23**

Further Mathematics

Advanced Subsidiary
Further Mathematics options
23: Further Statistics 1
(Part of options B, E, F and G)

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Advice

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Turn over

Pearson

1. The discrete random variable X has the following distribution

x	0	1	2	3	4
$\mathrm{P}(X = x)$	r	k	$\frac{k}{2}$	$\frac{k}{3}$	$\frac{k}{4}$

where r and k are positive constants.

The standard deviation of X equals the mean of X

Find the exact value of r

(6)

Question 1 continued

(Total for Question 1 is 6 marks)

2. A bag contains a large number of balls, all of the same size and weight. The balls are coloured Red, Blue or Yellow.

Jasmine asks each child in a group of 150 children to close their eyes, select a ball from the bag and show it to her. The child then replaces the ball and repeats the process a second time.

If both balls are the same colour the child receives a prize.

The results are given in the table below.

2nd colour \ 1st colour	Red	Blue	Yellow	Total
Red	31	11	18	60
Blue	8	10	9	27
Yellow	21	9	33	63
Total	60	30	60	150

Jasmine carries out a test, at the 5% level of significance, to see whether or not the colour of the 2nd ball is independent of the colour of the 1st ball.

(a) Calculate the expected frequencies for the cases where both balls are the same colour.

(2)

The test statistic Jasmine obtained was 12.712 to three decimal places.

(b) Use this value to complete the test, stating the critical value and conclusion clearly.

(3)

With reference to your calculations in part (a) and the nature of the experiment,

(c) give a plausible reason why Jasmine may have obtained her conclusion in part (b).

(1)

Question 2 continued

(Total for Question 2 is 6 marks)

3. A machine produces cloth. Faults occur randomly in the cloth at a rate of 0.4 per square metre.

The machine is used to produce tablecloths, each of area A square metres. One of these tablecloths is taken at random.

The probability that this tablecloth has no faults is 0.0907

(a) Find the value of A

(3)

The tablecloths are sold in packets of 20

A randomly selected packet is taken.

(b) Find the probability that more than 1 of the tablecloths in this packet has no faults.

(3)

A hotel places an order for 100 tablecloths each of area A square metres.

The random variable X represents the number of these tablecloths that have no faults.

(c) Find

(i) $\mathrm{E}(X)$

(ii) $\mathrm{Var}(X)$

(3)

(d) Use a Poisson approximation to estimate $\mathrm{P}(X = 10)$

(2)

It is claimed that a new machine produces cloth with a rate of faults that is less than 0.4 per square metre.

A piece of cloth produced by this new machine is taken at random.
The piece of cloth has area 30 square metres and is found to have 6 faults.

(e) Stating your hypotheses clearly, use a suitable test to assess the claim made for the new machine. Use a 5% level of significance.

(4)

(f) Write down the p-value for the test used in part (e).

(1)

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Question 3 continued

Question 3 continued

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Question 3 continued

(Total for Question 3 is 16 marks)

4. Table 1 below shows the number of car breakdowns in the *Snoreap* district in each of 60 months.

Number of car breakdowns	0	1	2	3	4	5
Frequency	12	11	19	14	3	1

Table 1

Anja believes that the number of car breakdowns per month in *Snoreap* can be modelled by a Poisson distribution. Table 2 below shows the results of some of her calculations.

Number of car breakdowns	0	1	2	3	4	$\geqslant 5$
Observed frequency (O_i)	12	11	19	14	3	1
Expected frequency (E_i)	9.92			9.64	4.34	

Table 2

(a) State suitable hypotheses for a test to investigate Anja's belief.
(1)

(b) Explain why Anja has changed the label of the final column to $\geqslant 5$
(1)

(c) Showing your working clearly, complete Table 2
(4)

(d) Find the value of $\frac{(O_i - E_i)^2}{E_i}$ when the number of car breakdowns is

(i) 1

(ii) 3
(2)

(e) Explain why Anja used 3 degrees of freedom for her test.
(2)

The test statistic for Anja's test is 6.54 to 2 decimal places.

(f) Stating the critical value and using a 5% level of significance, complete Anja's test.
(2)

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Question 4 continued

Question 4 continued

(Total for Question 4 is 12 marks)

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Turn over ▶

Pearson

1. Stuart is investigating a treatment for a disease that affects fruit trees. He has 400 fruit trees and applies the treatment to a random sample of these trees. The remainder of the trees have no treatment. He records the number of years, y, that each fruit tree remains free from this disease.

The results are summarised in the table below.

		Treatment	
		Applied	**Not applied**
Number of years free from this disease	$y < 1$	15	25
	$1 \leqslant y < 2$	35	61
	$2 \leqslant y$	124	140

The data are to be used to determine whether or not there is an association between the application of the treatment and the number of years that a fruit tree remains free from this disease.

(a) Calculate the expected frequencies for

(i) Applied and $y < 1$

(ii) Not applied and $1 \leqslant y < 2$

(2)

The value of $\sum \frac{(O-E)^2}{E}$ for the **other four** classes is 2.642 to 3 decimal places.

(b) Test, at the 5% level of significance, whether or not there is an association between the application of the treatment and the number of years a fruit tree remains free from this disease.

You should state your hypotheses, test statistic, critical value and conclusion clearly.

(5)

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Question 1 continued

Question 1 continued

Question 1 continued

(Total for Question 1 is 7 marks)

2. Xena catches fish at random, at a constant rate of 0.6 per hour.

(a) Find the probability that Xena catches exactly 4 fish in a 5-hour period.

(2)

The probability of Xena catching no fish in a period of t hours is less than 0.16

(b) Find the minimum value of t, giving your answer to one decimal place.

(3)

Independently of Xena, Zion catches fish at random with a mean rate of 0.8 per hour.

Xena and Zion try using new bait to catch fish. The number of fish caught in total by Xena and Zion after using the new bait, in a randomly selected 4-hour period, is 12

(c) Use a suitable test to determine, at the 5% level of significance, whether or not there is evidence that the rate at which fish are caught has increased after using the new bait. State your hypotheses clearly and the p-value used in your test.

(5)

Question 2 continued

Question 2 continued

Question 2 continued

(Total for Question 2 is 10 marks)

3. In a game, a coin is spun 5 times and the number of heads obtained is recorded.
Tao suggests playing the game 20 times and carrying out a chi-squared test to investigate whether the coin might be biased.

(a) Explain why playing the game only 20 times may cause problems when carrying out the test.
(1)

Chris decides to play the game 500 times. The results are as follows

Number of heads	0	1	2	3	4	5
Observed frequency	2	27	93	181	146	51

Chris decides to test whether or not the data can be modelled by a binomial distribution, with the probability of a head on each spin being 0.6

She calculates the expected frequencies, to 2 decimal places, as follows

Number of heads	0	1	2	3	4	5
Expected frequency	5.12	38.40	115.20	172.80	129.60	38.88

(b) State the number of degrees of freedom in Chris' test, giving a reason for your answer.
(1)

(c) Carry out the test at the 5% level of significance.
You should state your hypotheses, test statistic, critical value and conclusion clearly.
(5)

(d) Showing your working, find an alternative model which would better fit Chris' data.
(2)

Question 3 continued

Question 3 continued

Question 3 continued

(Total for Question 3 is 9 marks)

4. The discrete random variable X has the following probability distribution

x	0	2	3	6
$\mathrm{P}(X = x)$	p	0.25	q	0.4

(a) Find in terms of q

(i) $\mathrm{E}(X)$

(ii) $\mathrm{E}(X^2)$

(2)

Given that $\mathrm{Var}(X) = 3.66$

(b) show that $q = 0.3$

(3)

In a game, the score is given by the discrete random variable X

Given that games are independent,

(c) calculate the probability that after the 4th game has been played, the total score is exactly 20

(3)

A round consists of 4 games plus 2 bonus games. The bonus games are only played if after the 4th game has been played the total score is exactly 20

A prize of £10 is awarded if 6 games are played in a round **and** the total score for the round is at least 27

Bobby plays 3 rounds.

(d) Find the probability that Bobby wins at least £10

(6)

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Question 4 continued

Question 4 continued

(Total for Question 4 is 14 marks)

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- Try to answer every question.
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- Good luck with your examination.

Turn over

1. *Flobee* sells tomato seeds in packets, each containing 40 seeds. *Flobee* advertises that only 4% of its tomato seeds do not germinate.

Amodita is investigating the germination of *Flobee*'s tomato seeds. She plants 125 packets of *Flobee*'s tomato seeds and records the number of seeds that do not germinate in each packet.

Number of seeds that do not germinate	0	1	2	3	4	5	6 or more
Frequency	15	35	38	22	10	5	0

Amodita wants to test whether the binomial distribution B(40, 0.04) is a suitable model for these data.

The table below shows the expected frequencies, to 2 decimal places, using this model.

Number of seeds that do not germinate	0	1	2	3	4	5 or more
Expected Frequency	24.42	40.70	r	17.45	6.73	s

(a) Calculate the value of r and the value of s

(2)

(b) Stating your hypotheses clearly, carry out the test at the 5% level of significance. You should state the number of degrees of freedom, critical value and conclusion clearly.

(6)

Amodita believes that *Flobee* should use a more realistic value for the percentage of their tomato seeds that do not germinate.
She decides to test the data using a new model B(40, p)

(c) Showing your working, suggest a more realistic value for p

(2)

Question 1 continued

Question 1 continued

Question 1 continued

(Total for Question 1 is 10 marks)

2. Rowan and Alex are both check-in assistants for the same airline.
The number of passengers, R, checked in by Rowan during a 30-minute period can be modelled by a Poisson distribution with mean 28

(a) Calculate $\mathrm{P}(R \geqslant 23)$

(1)

The number of passengers, A, checked in by Alex during a 30-minute period can be modelled by a Poisson distribution with mean 16, where R and A are independent.
A randomly selected 30-minute period is chosen.

(b) Calculate the probability that exactly 42 passengers in total are checked in by Rowan and Alex.

(2)

The company manager is investigating the rate at which passengers are checked in.
He randomly selects 150 non-overlapping 60-minute periods and records the total number of passengers checked in by Rowan and Alex, in each of these 60-minute periods.

(c) Using a Poisson approximation, find the probability that for at least 25 of these 60-minute periods Rowan and Alex check in a total of fewer than 80 passengers.

(4)

On a particular day, Alex complains to the manager that the check-in system is working slower than normal. To see if the complaint is valid the manager takes a random 90-minute period and finds that the total number of people **Rowan** checks in is 67

(d) Test, at the 5% level of significance, whether or not there is evidence that the system is working slower than normal. You should state your hypotheses and conclusion clearly and show your working.

(4)

Question 2 continued

Question 2 continued

Question 2 continued

(Total for Question 2 is 11 marks)

3. The discrete random variable X has probability distribution

x	-3	-2	-1	0	2	5
$\mathrm{P}(X = x)$	0.3	0.15	0.1	0.15	0.1	0.2

(a) Find $\mathrm{E}(X)$

(1)

Given that $\mathrm{Var}(X) = 8.79$

(b) find $\mathrm{E}(X^2)$

(2)

The discrete random variable Y has probability distribution

y	-2	-1	0	1	2
$\mathrm{P}(Y = y)$	$3a$	a	b	a	c

where a, b and c are constants.

For the random variable Y

$$\mathrm{P}(Y \leqslant 0) = 0.75 \quad \text{and} \quad \mathrm{E}(Y^2 + 3) = 5$$

(c) Find the value of a, the value of b and the value of c

(5)

The random variable $W = Y - X$ where Y and X are independent.

The random variable $T = 3W - 8$

(d) Calculate $\mathrm{P}(W > T)$

(4)

Question 3 continued

Question 3 continued

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Question 3 continued

(Total for Question 3 is 12 marks)

4. Charlie carried out a survey on the main type of investment people have.
The contingency table below shows the results of a survey of a random sample of people.

		Main type of investment		
		Bonds	Cash	Stocks
Age	25–44	a	$b - e$	e
	45–75	c	$d - 59$	59

(a) Find an expression, in terms of a, b, c and d, for the difference between the observed and the expected value $(O - E)$ for the group whose main type of investment is Bonds and are aged 45–75
Express your answer as a single fraction in its simplest form.

(4)

Given that $\sum \frac{(O - E)^2}{E} = 9.62$ for this information,

(b) test, at the 5% level of significance, whether or not there is evidence of an association between the age of a person and the main type of investment they have.
You should state your hypotheses, critical value and conclusion clearly.
You may assume that no cells need to be combined.

(3)

Question 4 continued

Question 4 continued

(Total for Question 4 is 7 marks)

TOTAL FOR FURTHER STATISTICS 1 IS 40 MARKS

Please check the examination details below before entering your candidate information

Candidate surname	Other names

Pearson Edexcel Level 3 GCE

Centre Number | Candidate Number

Thursday 08 October 2020

Afternoon | Paper Reference **8FM0/23**

Further Mathematics

Advanced Subsidiary
Further Mathematics options
23: Further Statistics 1
(Part of options B, E, F and G)

You must have:
Mathematical Formulae and Statistical Tables (Green), calculator

Total Marks

Candidates may use any calculator allowed by Pearson regulations. Calculators must not have the facility for symbolic algebra manipulation, differentiation and integration, or have retrievable mathematical formulae stored in them.

Instructions

- Use **black** ink or ball-point pen.
- If pencil is used for diagrams/sketches/graphs it must be dark (HB or B).
- **Fill in the boxes** at the top of this page with your name, centre number and candidate number.
- Answer **all** questions and ensure that your answers to parts of questions are clearly labelled.
- Answer the questions in the spaces provided – *there may be more space than you need*.
- You should show sufficient working to make your methods clear. Answers without working may not gain full credit.
- Values from the statistical tables should be quoted in full. If a calculator is used instead of the tables, the value should be given to an equivalent degree of accuracy.
- Inexact answers should be given to three significant figures unless otherwise stated.

Information

- A booklet 'Mathematical Formulae and Statistical Tables' is provided.
- The total mark for this part of the examination is 40. There are 4 questions.
- The marks for **each** question are shown in brackets – *use this as a guide as to how much time to spend on each question.*

Advice

- Read each question carefully before you start to answer it.
- Try to answer every question.
- Check your answers if you have time at the end.

Turn over ▶

Pearson

1. A plumbing company receives call-outs during the working day at an average rate of 2.4 per hour.

(a) Find the probability that the company receives exactly 7 call-outs in a randomly selected 3-hour period of a working day.

(2)

The company has enough staff to respond to 28 call-outs in an 8-hour working day.

(b) Show that the probability that the company receives more than 28 call-outs in a randomly selected 8-hour working day is 0.022 to 3 decimal places.

(2)

In a random sample of 100 working days each of 8 hours,

(c) (i) find the expected number of days that the company receives more than 28 call-outs,

(1)

(ii) find the standard deviation of the number of days that the company receives more than 28 call-outs,

(2)

(iii) use a Poisson approximation to estimate the probability that the company receives more than 28 call-outs on at least 6 of these days.

(3)

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Question 1 continued

(Total for Question 1 is 10 marks)

2. In an experiment, James flips a coin 3 times and records the number of heads. He carries out the experiment 100 times with his left hand and 100 times with his right hand.

	Number of heads			
	0	**1**	**2**	**3**
Left hand	7	29	42	22
Right hand	13	35	36	16

(a) Test, at the 5% level of significance, whether or not there is an association between the hand he flips the coin with and the number of heads.

You should state your hypotheses, the degrees of freedom and the critical value used for this test.

(7)

(b) Assuming the coin is unbiased, write down the distribution of the number of heads in 3 flips.

(1)

(c) Carry out a χ^2 test, at the 10% level of significance, to test whether or not the distribution you wrote down in part (b) is a suitable model for the number of heads obtained in the 200 trials of James' experiment.

You should state your hypotheses, the degrees of freedom and the critical value used for this test.

(7)

Question 2 continued

(Total for Question 2 is 15 marks)

3. The probability distribution of the discrete random variable X is

$$P(X = x) = \begin{cases} \dfrac{k}{x} & \text{for } x = 1, 2 \text{ and } 3 \\ \dfrac{m}{2x} & \text{for } x = 6 \text{ and } 9 \\ 0 & \text{otherwise} \end{cases}$$

where k and m are positive constants.

Given that $E(X) = 3.8$, find $Var(X)$

(7)

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Question 3 continued

Question 3 continued

Question 3 continued

(Total for Question 3 is 7 marks)

4. During the morning, the number of cyclists passing a particular point on a cycle path in a 10-minute interval travelling eastbound can be modelled by a Poisson distribution with mean 8

The number of cyclists passing the same point in a 10-minute interval travelling westbound can be modelled by a Poisson distribution with mean 3

(a) Suggest a model for the total number of cyclists passing the point on the cycle path in a 10-minute interval, stating a necessary assumption.

(2)

Given that exactly 12 cyclists pass the point in a 10-minute interval,

(b) find the probability that at least 11 are travelling eastbound.

(3)

After some roadworks were completed, the total number of cyclists passing the point in a randomly selected 20-minute interval one morning is found to be 14

(c) Test, at the 5% level of significance, whether there is evidence of a decrease in the rate of cyclists passing the point.
State your hypotheses clearly.

(3)

Question 4 continued

Question 4 continued

(Total for Question 4 is 8 marks)

TOTAL FOR FURTHER STATISTICS 1 IS 40 MARKS

Please check the examination details below before entering your candidate information

Candidate surname	Other names

Pearson Edexcel Level 3 GCE

Centre Number

Candidate Number

Thursday 16 May 2019

Afternoon

Paper Reference **8FM0-23**

Further Mathematics
Advanced Subsidiary
Further Mathematics options
23: Further Statistics 1
(Part of options B, E, F and G)

You must have:
Mathematical Formulae and Statistical Tables (Green), calculator

Total Marks

Candidates may use any calculator allowed by Pearson regulations. Calculators must not have the facility for symbolic algebra manipulation, differentiation and integration, or have retrievable mathematical formulae stored in them.

Instructions

- Use **black** ink or ball-point pen.
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- Answer the questions in the spaces provided – *there may be more space than you need.*
- You should show sufficient working to make your methods clear. Answers without working may not gain full credit.
- Answers should be given to three significant figures unless otherwise stated.

Information

- A booklet 'Mathematical Formulae and Statistical Tables' is provided.
- The total mark for this part of the examination is 40. There are 4 questions.
- The marks for **each** question are shown in brackets – *use this as a guide as to how much time to spend on each question.*

Advice

- Read each question carefully before you start to answer it.
- Try to answer every question.
- Check your answers if you have time at the end.

Turn over ▶

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P 6 1 8 6 7 A 0 1 1 2

Pearson

1. A leisure club offers a choice of one of three activities to its 150 members on a Tuesday evening. The manager believes that there may be an association between the choice of activity and the age of the member and collected the following data.

Activity / Age a years	Badminton	Bowls	Snooker
$a < 20$	9	3	3
$20 \leqslant a < 40$	10	10	14
$40 \leqslant a < 50$	16	15	5
$50 \leqslant a < 60$	15	13	11
$a \geqslant 60$	4	19	3

(a) Write down suitable hypotheses for a test of the manager's belief.

(1)

The manager calculated expected frequencies to use in the test.

(b) Calculate the expected frequency of members aged 60 or over who choose snooker, used by the manager.

(1)

(c) Explain why there are 6 degrees of freedom used in this test.

(2)

The test statistic used to test the manager's belief is 19.583

(d) Using a 5% level of significance, complete the test of the manager's belief.

(2)

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Question 1 continued

(Total for Question 1 is 6 marks)

2. A spinner used for a game is designed to give scores with the following probabilities

Score	1	2	3	4	6
Probability	$\frac{3}{10}$	$\frac{1}{10}$	$\frac{1}{10}$	$\frac{2}{5}$	$\frac{1}{10}$

The spinner is spun 80 times and the results are as follows

Score	1	2	3	4	6
Frequency	15	4	12	41	8

Test, at the 10% level of significance, whether or not the spinner is giving scores as it is designed to do. Show your working and state your hypotheses clearly.

(7)

Question 2 continued

(Total for Question 2 is 7 marks)

3. Andreia's secretary makes random errors in his work at an average rate of 1.7 errors every 100 words.

(a) Find the probability that the secretary makes fewer than 2 errors in the next 100-word piece of work.

(2)

Andreia asks the secretary to produce a 250-word article for a magazine.

(b) Find the probability that there are exactly 5 errors in this article.

(2)

Andreia offers the secretary a choice of one of two bonus schemes, based on a random sample of 40 pieces of work each consisting of 100 words.

In scheme **A** the secretary will receive the bonus if more than 10 of the 40 pieces of work contain no errors.

In scheme **B** the bonus is awarded if the total number of errors in all 40 pieces of work is fewer than 56

(c) Showing your calculations clearly, explain which bonus scheme you would advise the secretary to choose.

(5)

Following the bonus scheme, Andreia randomly selects a single 500-word piece of work from the secretary to test if there is any evidence that the secretary's rate of errors has decreased.

(d) Stating your hypotheses clearly and using a 5% level of significance, find the critical region for this test.

(4)

Question 3 continued

Question 3 continued

Question 3 continued

(Total for Question 3 is 13 marks)

4. The discrete random variable X has probability distribution

x	-3	-1	1	2	4
$P(X = x)$	q	$\frac{7}{30}$	$\frac{7}{30}$	q	r

where q and r are probabilities.

(a) Write down, in terms of q, $P(X \leqslant 0)$ **(1)**

(b) Show that $E(X^2) = \frac{7}{15} + 13q + 16r$ **(2)**

Given that $E(X^3) = E(X^2) + E(6X)$

(c) find the value of q and the value of r **(7)**

(d) Hence find $P(X^3 > X^2 + 6X)$ **(4)**

Question 4 continued

Question 4 continued

(Total for Question 4 is 14 marks)

TOTAL FOR FURTHER STATISTICS 1 IS 40 MARKS

Write your name here

Surname	Other names

Pearson Edexcel
Level 3 GCE

Centre Number

Candidate Number

Further Mathematics

Advanced Subsidiary
Further Mathematics options
23: Further Statistics 1
(Part of options B, E, F and G)

Thursday 17 May 2018 – Afternoon

Paper Reference
8FM0-23

You must have:
Mathematical Formulae and Statistical Tables, calculator

Total Marks

Candidates may use any calculator allowed by the regulations of the Joint Council for Qualifications. Calculators must not have the facility for symbolic algebra manipulation, differentiation and integration, or have retrievable mathematical formulae stored in them.

Instructions

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- Answer the questions in the spaces provided – *there may be more space than you need.*
- You should show sufficient working to make your methods clear. Answers without working may not gain full credit.
- Answers should be given to three significant figures unless otherwise stated.

Information

- A booklet 'Mathematical Formulae and Statistical Tables' is provided.
- The total mark for this part of the examination is 40. There are 4 questions.
- The marks for **each** question are shown in brackets – *use this as a guide as to how much time to spend on each question.*

Advice

- Read each question carefully before you start to answer it.
- Try to answer every question.
- Check your answers if you have time at the end.

Turn over

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P 6 0 1 5 4 A 0 1 1 6

Pearson

Answer ALL questions. Write your answers in the spaces provided.

1. A researcher is investigating the distribution of orchids in a field. He believes that the Poisson distribution with a mean of 1.75 may be a good model for the number of orchids in each square metre. He randomly selects 150 non-overlapping areas, each of one square metre, and counts the number of orchids present in each square.

The results are recorded in the table below.

Number of orchids in each square metre	0	1	2	3	4	5	6
Number of squares	30	42	35	26	11	6	0

He calculates the **expected** frequencies as follows

Number of orchids in each square metre	0	1	2	3	4	5	More than 5
Number of squares	26.07	45.62	39.91	23.28	10.19	3.57	r

(a) Find the value of r giving your answer to 2 decimal places.

(1)

The researcher will test, at the 5% level of significance, whether or not the data can be modelled by a Poisson distribution with mean 1.75

(b) State clearly the hypotheses required to test whether or not this Poisson distribution is a suitable model for these data.

(1)

The test statistic for this test is 2.0 and the number of degrees of freedom to be used is 4

(c) Explain fully why there are 4 degrees of freedom.

(2)

(d) Stating your critical value clearly, determine whether or not these data support the researcher's belief.

(2)

The researcher works in another field where the number of orchids in each square metre is known to have a Poisson distribution with mean 1.5

He randomly selects 200 non-overlapping areas, each of one square metre, in this second field, and counts the number of orchids present in each square.

(e) Using a Poisson approximation, show that the probability that he finds at least one square with exactly 6 orchids in it is 0.506 to 3 decimal places.

(4)

Question 1 continued

Question 1 continued

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Question 1 continued

(Total for Question 1 is 10 marks)

2. The number of heaters, H, bought during one day from *Warmup* supermarket can be modelled by a Poisson distribution with mean 0.7

(a) Calculate $P(H \geqslant 2)$

(1)

The number of heaters, G, bought during one day from *Pumraw* supermarket can be modelled by a Poisson distribution with mean 3, where G and H are independent.

(b) Show that the probability that a total of fewer than 4 heaters are bought from these two supermarkets in a day is 0.494 to 3 decimal places.

(2)

(c) Calculate the probability that a total of fewer than 4 heaters are bought from these two supermarkets on at least 5 out of 6 randomly chosen days.

(3)

December was particularly cold. Two days in December were selected at random and the total number of heaters bought from these two supermarkets was found to be 14

(d) Test whether or not the mean of the total number of heaters bought from these two supermarkets had increased. Use a 5% level of significance and state your hypotheses clearly.

(5)

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Question 2 continued

Question 2 continued

Question 2 continued

(Total for Question 2 is 11 marks)

3. A fair six-sided black die has faces numbered 1, 2, 2, 3, 3 and 4

The random variable B represents the score when the black die is rolled.

(a) Write down the value of $\mathrm{E}(B)$

(1)

A white die has 6 faces numbered 1, 1, 2, 4, 5 and c where $c > 5$
The discrete random variable W represents the score when the white die is rolled and has probability distribution given by

w	1	2	4	5	c
$\mathrm{P}(W = w)$	$a + b$	a	0.3	a	b

Greg and Nilaya play a game with these dice.

Greg throws the black die and Nilaya throws the white die. Greg wins the game if he scores at least two more than Nilaya, otherwise Greg loses.

The probability of Greg winning the game is $\frac{1}{6}$

(b) Find the value of a and the value of b
Show your working clearly.

(5)

The random variable $X = 2W - 5$

Given that $\mathrm{E}(X) = 2.6$

(c) find the exact value of $\mathrm{Var}(X)$

(6)

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Question 3 continued

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Question 3 continued

(Total for Question 3 is 12 marks)

Turn over

4. Abram carried out a survey of two treatments for a plant fungus. The contingency table below shows the results of a survey of a random sample of 125 plants with the fungus.

		Treatment		
		No action	Plant sprayed once	Plant sprayed every day
Outcome	Plant died within a month	15	16	25
	Plant survived for 1 – 6 months	8	25	10
	Plant survived beyond 6 months	7	14	5

Abram calculates expected frequencies to carry out a suitable test. Seven of these are given in the partly-completed table below.

		Treatment		
		No action	Plant sprayed once	Plant sprayed every day
Outcome	Plant died within a month			17.92
	Plant survived for 1 – 6 months	10.32	18.92	13.76
	Plant survived beyond 6 months	6.24	11.44	8.32

The value of $\sum \frac{(O-E)^2}{E}$ for the 7 given values is 8.29

Test at the 2.5% level of significance, whether or not there is an association between the treatment of the plants and their survival. State your hypotheses and conclusion clearly.

(7)

Question 4 continued

Question 4 continued

(Total for Question 4 is 7 marks)

TOTAL FOR FURTHER STATISTICS 1 IS 40 MARKS

Write your name here

Surname	Other names

Pearson Edexcel Level 3 GCE

Centre Number	Candidate Number

Further Mathematics

Advanced Subsidiary
Further Mathematics options
Paper 2B: Further Pure Mathematics 1 and Further Statistics 1

Sample Assessment Material for first teaching September 2017
Time: 1 hour 40 minutes

Paper Reference
8FM0/2B

You must have:
Mathematical Formulae and Statistical Tables, calculator

Total Marks

Candidates may use any calculator permitted by Pearson regulations. Calculators must not have the facility for algebraic manipulation, differentiation and integration, or have retrievable mathematical formulae stored in them.

Instructions

- Use **black** ink or ball-point pen.
- If pencil is used for diagrams/sketches/graphs it must be dark (HB or B).
- **Fill in the boxes** at the top of this page with your name, centre number and candidate number.
- There are **two** sections in this question paper. Answer **all** the questions in Section A and **all** the questions in Section B.
- Answer the questions in the spaces provided – *there may be more space than you need.*
- You should show sufficient working to make your methods clear. Answers without working may not gain full credit.
- Answers should be given to three significant figures unless otherwise stated.

Information

- A booklet 'Mathematical Formulae and Statistical Tables' is provided.
- There are 9 questions in this question paper. The total mark for this paper is 80.
- The marks for **each** question are shown in brackets – *use this as a guide as to how much time to spend on each question.*

Advice

- Read each question carefully before you start to answer it.
- Try to answer every question.
- Check your answers if you have time at the end.

Turn over ▶

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Pearson

SECTION B

Answer ALL questions. Write your answers in the spaces provided.

6. A university foreign language department carried out a survey of prospective students to find out which of three languages they were most interested in studying.

A random sample of 150 prospective students gave the following results.

		Language		
		French	**Spanish**	**Mandarin**
Gender	**Male**	23	22	20
	Female	38	32	15

A test is carried out at the 1% level of significance to determine whether or not there is an association between gender and choice of language.

(a) State the null hypothesis for this test.
(1)

(b) Show that the expected frequency for females choosing Spanish is 30.6
(1)

(c) Calculate the test statistic for this test, stating the expected frequencies you have used.
(3)

(d) State whether or not the null hypothesis is rejected. Justify your answer.
(2)

(e) Explain whether or not the null hypothesis would be rejected if the test was carried out at the 10% level of significance.
(1)

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Question 6 continued

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Question 6 continued

Question 6 continued

(Total for Question 6 is 8 marks)

7. The discrete random variable X has probability distribution given by

x	-1	0	1	2	3
$P(X = x)$	c	a	a	b	c

The random variable $Y = 2 - 5X$

Given that $E(Y) = -4$ and $P(Y \geqslant -3) = 0.45$

(a) find the probability distribution of X.

(7)

Given also that $E(Y^2) = 75$

(b) find the exact value of $\text{Var}(X)$

(2)

(c) Find $P(Y > X)$

(2)

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Question 7 continued

Question 7 continued

Question 7 continued

(Total for Question 7 is 11 marks)

8. Two car hire companies hire cars independently of each other.

Car Hire A hires cars at a rate of 2.6 cars per hour.

Car Hire B hires cars at a rate of 1.2 cars per hour.

(a) In a 1 hour period, find the probability that each company hires exactly 2 cars.

(2)

(b) In a 1 hour period, find the probability that the total number of cars hired by the two companies is 3

(2)

(c) In a 2 hour period, find the probability that the total number of cars hired by the two companies is less than 9

(2)

On average, 1 in 250 new cars produced at a factory has a defect.

In a random sample of 600 new cars produced at the factory,

(d) (i) find the mean of the number of cars with a defect,

(ii) find the variance of the number of cars with a defect.

(2)

(e) (i) Use a Poisson approximation to find the probability that no more than 4 of the cars in the sample have a defect.

(ii) Give a reason to support the use of a Poisson approximation.

(2)

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Question 8 continued

Question 8 continued

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Question 8 continued

(Total for Question 8 is 10 marks)

9. The discrete random variable X follows a Poisson distribution with mean 1.4

(a) Write down the value of

(i) $P(X = 1)$

(ii) $P(X \leqslant 4)$

(2)

The manager of a bank recorded the number of mortgages approved each week over a 40 week period.

Number of mortgages approved	0	1	2	3	4	5	6
Frequency	10	16	7	4	2	0	1

(b) Show that the mean number of mortgages approved over the 40 week period is 1.4

(1)

The bank manager believes that the Poisson distribution may be a good model for the number of mortgages approved each week.

She uses a Poisson distribution with a mean of 1.4 to calculate expected frequencies as follows.

Number of mortgages approved	0	1	2	3	4	5 or more
Expected frequency	9.86	r	9.67	4.51	1.58	s

(c) Find the value of r and the value of s giving your answers to 2 decimal places.

(2)

The bank manager will test, at the 5% level of significance, whether or not the data can be modelled by a Poisson distribution.

(d) Calculate the test statistic and state the conclusion for this test. State clearly the degrees of freedom and the hypotheses used in the test.

(6)

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Question 9 continued

Question 9 continued

Question 9 continued

(Total for Question 9 is 11 marks)

TOTAL FOR SECTION B IS 40 MARKS
TOTAL FOR PAPER IS 80 MARKS

Please check the examination details below before entering your candidate information

Candidate surname	Other names

Pearson Edexcel Level 3 GCE

Centre Number | Candidate Number

Specimen Paper

Paper Reference **8FM0/23**

Further Mathematics
Advanced Subsidiary
23: Further Statistics 1

You must have:
Mathematical Formulae and Statistical Tables, calculator

Total Marks

Candidates may use any calculator allowed by the regulations of the Joint Council for Qualifications. Calculators must not have the facility for symbolic algebra manipulation, differentiation and integration, or have retrievable mathematical formulae stored in them.

Instructions
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- If pencil is used for diagrams/sketches/graphs it must be dark (HB or B).
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- Answer **all** questions and ensure that your answers to parts of questions are clearly labelled.
- Answer the questions in the spaces provided – *there may be more space than you need.*
- You should show sufficient working to make your methods clear. Answers without working may not gain full credit.
- Answers should be given to three significant figures unless otherwise stated.

Information
- A booklet 'Mathematical Formulae and Statistical Tables' is provided.
- The total mark for this part of the examination is 40. There are 4 questions.
- The marks for **each** question are shown in brackets – *use this as a guide as to how much time to spend on each question.*

Advice
- Read each question carefully before you start to answer it.
- Try to answer every question.
- Check your answers if you have time at the end.

Turn over ▶

Answer ALL questions. Write your answers in the spaces provided.

1. In a survey, 200 people aged 25 and older were randomly selected and asked how much time they spend on social media each day. The table below shows a summary of the results.

		Less than 1 hour	1 hour or more
Age	25 to 54	60	74
	55 and older	32	34

Noah carries out a test, at the 5% level of significance, to see if there is an association between age and time spent on social media. He uses the hypotheses

H_0: There is no association between age and time spent on social media.
H_1: There is an association between age and time spent on social media.

He calculates $\sum \frac{(O-E)^2}{E} = 0.245$ for this information.

(a) State the conclusion of the test. Justify your answer.
(2)

(b) Explain why having a large number of age groups may cause a problem when carrying out the hypothesis test.
(1)

Jade decides to take the same information and subdivide the age groups.

She then uses the information in the partially complete table below to carry out a test, at the 5% level of significance, of the same hypotheses.

		Observed	
		Less than 1 hour	1 hour or more
Age	25 to 34	18	28
	35 to 44	20	25
	45 to 54	22	21
	55 to 64	25	15
	65 and older	7	19

	$\frac{(O-E)^2}{E}$	
	Less than 1 hour	1 hour or more
25 to 34	0.47	0.40
35 to 44	0.02	0.02
45 to 54	0.25	0.21
55 to 64	2.37	2.02
65 and older		

(c) Complete Jade's hypothesis test. State clearly the degrees of freedom and the critical value used in the test.
(4)

(d) State, giving a reason, which of the conclusions in part (a) and part (c) you believe to be the more reliable.
(1)

Question 1 continued

Question 1 continued

Question 1 continued

(Total for Question 1 is 8 marks)

2. The discrete random variable X has probability distribution given by

x	2	3	6	11
$\mathrm{P}(X = x)$	$\frac{1}{50}$	a	$\frac{1}{25}$	b

The discrete random variable $Y = X^2$

Given that $\mathrm{E}(Y) = 50.3$

(a) find the value of a and the value of b

(3)

(b) Find $\mathrm{P}(9 - Y > 0)$

(2)

Independent observations $X_1, X_2, X_3, \ldots, X_{120}$ of X are taken.

The random variable T represents the total number of these 120 observations that are even.

(c) Find

(i) $\mathrm{E}(T)$

(ii) $\mathrm{Var}(T)$

(2)

(d) Find, using a suitable approximation, $\mathrm{P}(T > 10)$

(3)

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Question 2 continued

(Total for Question 2 is 10 marks)

3. A hotel has 30 rooms. The manager models the number of empty rooms each Friday night using a binomial distribution, B(30, 0.08)

The manager recorded the number of empty rooms in the hotel each Friday night over a period of 80 weeks.

Number of empty rooms	0	1	2	3	4	5	6 or more
Frequency	14	18	22	11	10	5	0

The table below shows the expected frequencies using the manager's model.

Number of empty rooms	0	1	2	3	4	5 or more
Expected frequency	6.56	17.11	r	17.50	10.27	s

(a) Find the value of r and the value of s

(2)

(b) Stating your hypotheses clearly, test the manager's model at the 5% level of significance.

(6)

(c) Suggest an improved model for the number of empty rooms in the hotel each Friday night.

(2)

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Question 3 continued

(Total for Question 3 is 10 marks)

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4. An office has a photocopier and a printer. The photocopier and the printer break down independently.

The number of breakdowns per month for the photocopier follows a Poisson distribution with mean 2.4

The number of breakdowns per month for the printer follows a Poisson distribution with mean 1.6

(a) Determine which machine is more likely to break down exactly twice in one month.
(2)

(b) Work out the probability that both machines break down at most once in one month.
(2)

In a randomly selected month there were 4 breakdowns.

(c) Find the probability that in this month the photocopier broke down more than the printer.
(4)

A repair company carried out repairs on the photocopier and the printer. Following the repairs, there were a total of 3 breakdowns in two months.

(d) Test, at the 5% level of significance, whether or not there is evidence that the rate of breakdowns has decreased following the repairs.
(4)

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Question 4 continued

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Question 4 continued

(Total for Question 4 is 12 marks)

TOTAL FOR FURTHER STATISTICS 1 IS 40 MARKS